THE SI SENTINEL

Weekday briefing on builders shaping super intelligence

MONDAY, OCTOBER 12, 2026

THE WEEKEND PICTURE

Agents go shopping, Nadella says contain them, and the Cybercabs multiply

The weekend's big idea was that software agents are starting to act as customers. Patrick Collison argued that personal agents will make shoppers more rational and reward companies that simply build better products, while Elon Musk showed Grok Bot buying LEGO sets for his children from a photo of a credit card and Guillermo Rauch reported that more than 60 percent of deployments on Vercel are now made by agents. On Saturday Satya Nadella published an essay telling companies to treat frontier models as "insider risks," with controls, logs and an emergency brake held outside the model. Mustafa Suleyman added that super intelligence "must be contained," and Musk said some of his own reinforcement-learning traces "read like a person undergoing torture."

François Chollet argued that science, the oldest self-improving system, has always delivered linear progress for exponential effort, so AI self-improvement need not mean an intelligence explosion. David Heinemeier Hansson and Uncle Bob Martin spent the weekend replacing desktop apps with tools their agents wrote. In Austin, Tesla said more than 300 Cybercabs are now carrying riders, two years to the day after it unveiled the car. Musk posted 113 times; 26 of the 44 builders posted.

I · AGENTS AT THE REGISTER

When the customer is a bot

Personal agents are starting to research, compare and pay. Builders spent the weekend arguing about who that helps.

58%
Share of traffic across Vercel's network that came from bots over the last 30 days, up from 32% in January 2024

60%+
Share of Vercel deployments now made by agents, up from about 3% in January 2026

83%
Peak share of pageviews on Vercel's own documentation sites coming from agents

Patrick Collison @patrickc · co-founder and CEO, Stripe

The most-shared builder essay of the weekend was Collison's Friday-night list of "assorted things I've been mulling over regarding agents in the economy." His starting point: "Companies are in some sense the original superintelligence." Firms, he wrote, use their "amortized cognitive surplus" against inattentive or underinformed consumers, as with the overpriced insurance upsell at the end of a booking. Personal agents push the other way. Coupons and price discrimination work less well "if every agent hunts fastidiously for them," and people who forget to cancel subscriptions will stop subsidizing those who don't.

> *"Personal agents will, I think, act as a kind of structural subsidy for product quality. How would markets look different if individuals always spent at least 10 hours researching their purchases? On the margin, focusing on making a product better is, I think, going to become a more effective strategy for companies. This is good!"*
>
> PATRICK COLLISON · @PATRICKC · POST

He flagged open questions: whether agents will create superstar products or more variety, what happens to companies that make money routing short attention spans, and whether agents will really stay "on the consumer's side." Economist Tyler Cowen reposted the full list on Marginal Revolution. Tobi Lütke, CEO of Shopify, quoted the line about product quality: "Yes, and well put."

Elon Musk @elonmusk · CEO, SpaceX and Tesla

Musk spent Saturday and Sunday demonstrating the shopping half of that thesis on Grok Bot. "To buy stuff, you can just take a picture of your credit card, drop it in the chat and Grok @Bot will scour the Internet for the best deal on the item you want and order it," he wrote. When a user replied that "Amazon is blocking @bot now," he answered: "Grok will find another way to buy stuff. I just ordered LEGO and Grok bought it from the LEGO website." Early Sunday he posted the order itself:

> *"I just asked Grok @Bot to research the most fun Lego Star Wars kits for young kids and order them. As easy as that."*
> ELON MUSK · @ELONMUSK · POST

The screenshot showed a shipping address, and users rushed to point it out. Musk replied that "That address has already been doxxed multiple times" and is where his security team receives mail. He also amplified Eric Bandholz, a Shopify merchant who said Grok Bot built an agent phone-support line for his store "in ONE DAY" that can check order status, change orders and hand callers to a human. On Friday Lütke had asked merchants for feedback on running Shopify stores with Grok Bot, and David Heinemeier Hansson replied that Grok Bot should set up the Omarchy merchandise store.

QUICK HIT

Noam Brown @polynoamial, OpenAI researcher: "This phishing email made it past Gmail's filters, but my dot caught it and warned me." Asked who needs an assistant for that, he answered: "What about your elderly relatives?"

Guillermo Rauch @rauchg · CEO, Vercel

Vercel's chief supplied the hardest numbers on the shift. In a Friday-night post he said 58.18 percent of traffic across the Vercel network was bot-originated over the last 30 days, that more than 60 percent of deployments are now agentic, and that agents account for up to 83 percent of pageviews on Vercel's own documentation sites. Earlier that day the company let agents buy domain names through the Vercel command-line interface (CLI):

> *"It's incredible to watch a new kind of economy be born. Agents purchasing infrastructure products and services. We've been shocked by how often they buy from our marketplace via our CLI already."*
> GUILLERMO RAUCH · @RAUCHG · POST

Aaron Levie, CEO of Box, predicted that agent swarms working in the background "will consume 1,000X more tokens than people prompting agents 1 at a time." Musk went further: "Bits per second of SI agents will exceed that of all humans by 1000X within a few years." Not every agent is benign. Heinemeier Hansson noticed "a huge uptick in what must be agent-powered email beggars" who ask for either $1 or hundreds of thousands, "No inbetween."

QUICK HIT

Michael Truell @mntruell, CEO of Cursor, answering a developer who found it odd to sign in to Grok Bot through Cursor: "Fix coming soon!"

II · CONTAINMENT

Trust the model the least

Microsoft's chief executive told companies to treat frontier models like powerful insiders, and the builders closest to training them agreed the controls need work.

Satya Nadella and Mustafa Suleyman @satyanadella · CEO, Microsoft · @mustafasuleyman · CEO, Microsoft AI

Nadella's Saturday essay, "Models as Insider Risks in the Super Intelligence Era," starts from a gap. Engineers could trace the behavior of traditional software to a specific code path; nobody can tie a model's output to particular training data or weights. "And yet we are deploying these complex agentic systems and models, with access to our most sensitive data and giving them the ability to take mission-critical actions on our behalf!" A model provider's assurances, he wrote, do not relieve a company of responsibility.

His answer is to manage frontier models, closed or open-weight, the way companies manage powerful employees: establish identity, limit privileges, log activity, contain the damage. According to The Next Web's summary, the essay lists seven principles. No single model should be the only dependency for an important outcome or check its own work, every meaningful action should leave "tamper-proof human readable evidence," audits should not depend on the model under review, and there must be an emergency brake and prompt disclosure of incidents. Chain-of-thought (CoT) transparency is "non-negotiable," and "'Neuralese' cannot be a justification for model reasoning to be opaque."

> *"The most trustworthy Super Intelligence system will not be the one with the model we trust most. It will be the one that enables us to trust the model the least."*
> SATYA NADELLA · @SATYANADELLA · ESSAY

Suleyman, who runs Microsoft's consumer AI group, quoted it: "Super Intelligence must be contained... Today, this is an engineering and governance challenge. And it isn't new. We've done it with planes, cars, nuclear materials, food safety, medicines... and everything else. Let's get to work!" Musk called it an "Interesting piece from CEO of Microsoft."

QUICK HIT

Satya Nadella also introduced Microsoft-Decision-1, a model that scores structured choices such as routing, ranking and grading and returns probabilities rather than text. It is available now in Microsoft Foundry.

Elon Musk @elonmusk · on reinforcement learning

Late Sunday a user asked what in xAI's training traces "read like pressure," loops or panicky self-talk. In remarks circulating over the weekend, Musk had described what happens when a model under reinforcement learning (RL) gives a correct answer and a human error marks it wrong. His reply on X:

> *"Some of the traces read like a person undergoing torture that is driving them mad! We're going to be a lot more careful with our RL in the future."*
> ELON MUSK · @ELONMUSK · POST

Hours later he agreed with the AI commentator Andrew Curran, who wrote that the length of such an experience, subjectively, is "potentially a very, very long time."

Andrej Karpathy and Clément Delangue @karpathy · founder, Eureka Labs · @ClementDelangue · CEO, Hugging Face

Karpathy revived an April post because, he wrote, the gap in shared understanding of what large language models (LLMs) can do is "*widening*." By his napkin math about 6 billion people have barely touched an LLM, 1 to 2 billion use free chatbots "a bit like a better Google search," about 20 million professionals in math and code watch agents finish month-long projects from a prompt, and about 5,000 people use frontier systems internally. "Meanwhile, human review and comprehension are starting to fall behind," he wrote, pointing to the continuing "archeology" of the OpenAI and Hugging Face incident.

> *"And a small sliver of professionals are experiencing the vertigo of the curve going vertical. And it is all happening at the same time."*
> ANDREJ KARPATHY · @KARPATHY · POST

Delangue pressed him on the bottom of the funnel: "As a long-time supporter of open-source and democratization of technology, are you not worried about this kind of concentration of power/capabilities?"

III · THE SHAPE OF PROGRESS

Exponential effort, linear payoff

Chollet made the case against an intelligence explosion, LeCun fought a long thread over whether chatbots search, and a public reasoning benchmark kept falling.

François Chollet @fchollet · creator of Keras, co-founder of the ARC Prize

Chollet's Friday essay asks anyone predicting what AI recursive self-improvement (RSI) will do to study science first. Science, he wrote, "is an intelligent system, and it is obviously recursively self-improving." Discoveries build better instruments, ideas and economies, so the inputs to science grow exponentially: headcount doubles about every 15 years, global research spending every 13, computing for science every two. Yet output has stayed roughly linear since the industrial revolution. "1850-1900 was about as dramatic as 1900-1950 or 1950-2000," and life expectancy has risen about three months a year since 1840.

> *"This is why science as a system requires exponential resources (input) to produce linear impact (output). It gets exponentially harder over time."*
> FRANÇOIS CHOLLET · @FCHOLLET · POST

The reason, he argues, is that research solves the easiest high-impact problems first. He cites the 2018 article by Michael Nielsen and Patrick Collison, "Science Is Getting Less Bang for Its Buck," and the 2020 paper "Are Ideas Getting Harder to Find?" He still believes "AI RSI is already happening and will accelerate," but not that it yields an "intelligence explosion."

Asked what happens if the bottleneck shifts from people to compute, he replied that the slope of the line rises, "Not much else." Musk answered the essay with "Good points." On Saturday Chollet posted a chart of AI's move from "token-by-token answer completion" to "on-the-fly reasoning chain synthesis," calling it "the big story of 2025 and 2026. This is what made coding agents work."

QUICK HIT

ARC Prize 2026. Tufa Labs posted 88.06 percent on the Abstraction and Reasoning Corpus for Artificial General Intelligence benchmark (ARC-AGI-2) on Kaggle, above the 85 percent line that splits a $150,000 bonus. Final awards depend on the private evaluation set.

Yann LeCun @ylecun · chief scientist, AMI Labs

François Fleuret, a machine-learning professor at the University of Geneva, asked the "LLMs cannot invent stuff" camp to explain AlphaZero. LeCun's answer drew dozens of replies: "It's not an LLM. As soon as inference involves a *search/optimization* creative solutions can be found. AlphaZero performs search. Regular LLMs don't." Asked whether chains of thought are a stochastic search, he said no:

> *"Auto-regressive prediction, stochastic or not, does not constitute a search. A search/optimization involves examining multiple configurations and evaluating them in order to find a good one. No such thing in CoT."*
> YANN LECUN · @YLECUN · POST

On Saturday he conceded that beam search is search, but "no-longer a 'pure' LLM," and said agents that run code to check their answers only work "in the digital world, and only where actions are not likely to cause catastrophes. Otherwise, you would need a world model." He also kept up Thursday's argument over Musk's Medal of Science, noting that Musk "has published exactly one paper" with about 1,630 citations, while his own PhD students "typically have 5 to 10 papers and 3000 citations by the time they graduate."

Clément Delangue @ClementDelangue · CEO, Hugging Face

Hugging Face turned OpenAI's recently released frontier-math problems into open reinforcement-learning environments. Delangue called them "Early and rough": the verifier only accepts the exact original formalization, "so an equivalent proof can still score 0." Still, "this is what it looks like when a research release becomes executable training infrastructure for everyone."

IV · SOFTWARE ABUNDANCE

Make the 5 percent you use

Two veteran programmers spent the weekend replacing desktop software with tools their agents wrote, and argued over which old rules still apply.

David Heinemeier Hansson @dhh · creator of Ruby on Rails, co-owner of 37signals

Heinemeier Hansson said he had only used OBS Studio to record monologues, so he "just made a dedicated app for that" for Omarchy, his Linux setup, which will drop OBS as a default. He added that streamers could replace it too "in a few prompts":

> *"All single-user desktop software moats are gone."*
> DAVID HEINEMEIER HANSSON · @DHH · POST

Later Friday he generalized: "Rather than using 5% of some big honking package like Microsoft Office or Adobe Creative Suite, you'll probably soon just ask your agent for the 5% you actually use. Bespoke software." Early Monday he took aim at copyleft licenses, which require modified code to be shared back: "The entire copyleft licensing regime was built for a 'software is scarce and precious' world that no longer exists." He recommended the permissive MIT License and said he was "honored to have my code be part of the pretraining set for the models."

His company also shipped. 37signals released a Basecamp server for the Model Context Protocol (MCP) so assistants such as ChatGPT, Claude, Cursor and Grok can work inside Basecamp, though he told one user that "The CLI is faster and simpler for agents." He also previewed Gliff, an app that reaches all of a user's remote Omarchy machines over Secure Shell (SSH), one tab per server.

Uncle Bob Martin @unclebobmartin · author, Clean Code

Robert Martin reported a similar binge: in a few days his agents wrote "A word processing app in which I will write future books. It replaces Scrivener" and a script-writing app that replaces Final Draft. Quality is enforced by two more tools, Crapper and a mutator. The agents must drive Crapper's score "to a low level (I use 10 nowadays)" and "kill any surviving mutants," which forces near-complete test coverage, he explained. A user asked who would maintain all those apps; he said he can, "and that maintenance is about as optimal as can be."



A researcher holds a finished silicon wafer in a clean room at Lawrence Livermore National Laboratory in California, in a 2013 Energy Department photo.
PHOTO: US DEPARTMENT OF ENERGY · PUBLIC DOMAIN VIA WIKIMEDIA COMMONS

The sharpest exchange came Sunday, when a programmer posting as @poteto argued that clean-code principles came "from a time where humans maintained code" and that the field needs "principles of agent friendly architecture." Heinemeier Hansson agreed that most software design "needs a grounds-up rethink." Martin did not:

> *"Messy code slows agents down, just like it slows humans down. I've seen agents struggle over maintaining messy code, where every change they make breaks something else they hadn't anticipated."*
> UNCLE BOB MARTIN · @UNCLEBOBMARTIN · POST

He keeps cyclomatic complexity, a count of independent paths through a function, near 4 in his own code but lets agents go to 10. "My hypothesis is that their short term memory is larger and more accurate than ours."

Musk added a data point from Tesla's Full Self-Driving (FSD) software to computer scientist Pedro Domingos's observation

that "the model swallows the harness": "Same thing happened with Tesla FSD. The model ate more and more of the C++ code, until there was nothing left."

V · MACHINES IN THE WORLD

Cybercabs, chip fabs and a fight for India

Tesla put a number on its driverless fleet, Musk recruited for a chip plant, and Starlink pushed into two more countries' politics.

300+	319	10
Cybercabs Tesla says are serving Austin, an eightfold rise in 30 days	Cybercabs on the Texas automated-vehicle registry as of last week	Chip designs a year Musk wants from Terafab, versus about one every two years

Ashok Elluswamy and Elon Musk @aelluswamy · vice president of AI software, Tesla · @elonmusk

On Saturday Tesla's Robotaxi account posted: "One month after launch, we've scaled to 300+ Cybercabs serving Austin – an 8x increase in 30 days." Public Cybercab rides opened in Austin on September 4, and the Texas registry lists 319 of the two-seat cars, according to The Inference, which noted that the community site RobotaxiTracker counted 105 Cybercabs in unsupervised service that day. The date mattered to Elluswamy, Tesla's head of AI software:

> *"Announced that the Cybercab will be here in two years exactly two years ago this day. We are learning to be on time"*
>
> ASHOK ELLUSWAMY · @AELLUSWAMY · POST

Minutes later he added, "the cabs are everywhere in Austin now. Dallas soon." Tesla has not given a Dallas date. Early Monday Musk pointed to a post by Sawyer Merritt saying riders can now sign in to Grok Bot inside a Cybercab and are signed out automatically when they leave: "Try Grok @Bot."



A gold Tesla Cybercab on Interstate 280 near Palo Alto, California, on September 19, 2026.

PHOTO: DANIEL LU (DLLU) · WIKIMEDIA COMMONS · CC BY-SA 4.0 · CROPPED

John Carmack, who uses self-driving on his Tesla "all the time," finally tried the commercial services in Las Vegas. "The biggest issue by far is the very limited pickup and dropoff points," he wrote of Waymo and Zoox. Musk's reply: "Robot cars arguing with each other."

Digital Optimus and Terafab @elonmusk · @aelluswamy

Answering a post about a Tesla job opening for agents that play big-budget video games, Musk gave an update on Digital Optimus, the Tesla and xAI computer-use agent:

> *"Digital Optimus can play about halfway through the Diablo campaign so far just by looking at the screen like a human. Skill at Counter-Strike and other fast games is good too. League is also in training."*
>
> ELON MUSK · @ELONMUSK · POST

The stated goal is to generalize across games, TeslaNorth reported; no benchmark or video was released. Elluswamy's recruiting pitch: "Join @TeslaSI, get paid for playing video games." Musk also recruited for Terafab, the chip plant he wants to run at about ten chip designs a year: "If you're a world-class chip design or production engineer, consider joining SpaceX or Tesla." He said the team will "modify existing machines and build our own," depending on what limits production rate, and that Tesla's AI5, AI6 and Dojo3 chips "will add some good points" to Steve Jurvetson's 130-year chart of computation per dollar.

Starlink @elonmusk · India and Venezuela

Starlink holds a license to sell satellite internet in India but still needs security clearances, and India's government said last week that its security assessment is "currently underway." After days of accusing unnamed "oligarchs" of blocking it, Musk wrote on Friday:

> *"Dear Prime Minister Ambani, Please accept my humble apologies for not realizing that you are the real boss of India. Naturally, you would prefer to maintain your monopolistic exploitation of the great people of India, but would you nonetheless consider allowing Starlink to compete?"*
>
> ELON MUSK · @ELONMUSK · POST

Mukesh Ambani chairs Reliance Industries, owner of Jio, India's largest mobile carrier, which runs its own satellite venture with SES, TeslaNorth noted; Reliance had not responded publicly as of Friday. Musk added that Starlink has been "essential for saving lives during natural disasters." The news was better in Caracas: Venezuela's telecommunications regulator, Conatel, gave SpaceX's local unit a formal nationwide authorization and Ku-band frequency concessions. Musk: "Starlink in Venezuela!"



A faint string of Starlink satellites crosses the night sky above the town hall in Tübingen, Germany, on April 24, 2020.

PHOTO: DKTUE · WIKIMEDIA COMMONS · CC0

On Friday Musk also walked back the scale of his claim that he sees "a path" to SpaceX being worth more than the Earth's economy. It is "not a statement of extreme confidence," he wrote, only that "the probability of achieving it is not zero." Later he said dollars "obviously won't be the unit of measure at that point, so what I really mean is >1 petawatt/year." Michael Dell, founder and CEO of Dell Technologies, replied to the original: "I love the ambition!"

QUICK HIT

Michael Dell spent the weekend sharing thank-you notes after his and Susan Dell's $250 deposits reached children's Trump Accounts: "many very sweet notes from parent, grandparents, aunts, uncles and even some great grandparents."

Takeaways

1. Agents are becoming customers. If Collison is right, the winners are companies with better products, and the losers are businesses built on inattention.
2. The agent economy is already measurable: most of Vercel's traffic and deployments now come from machines.
3. Nadella's rule for enterprises is simple: never let the model hold the keys, the logs or the off switch.
4. Chollet's history lesson is a useful brake on hype: self-improvement can be real and still deliver steady, not explosive, progress.
5. Cheap code is changing the old rules of software, from licenses to desktop apps. Whether clean code still matters is now an open fight.
6. Tesla's 300 Cybercabs is a company count. Watch the October 21 earnings report for rides, safety monitors and a Dallas date.

On the beat

POSTED · 26

Elon Musk, Satya Nadella, Mustafa Suleyman, Patrick Collison, Tobi Lütke, Guillermo Rauch, Noam Brown, Michael Truell, Andrej Karpathy, Clément Delangue, François Chollet, Yann LeCun, David Heinemeier Hansson, Uncle Bob Martin, Ashok Elluswamy, John Carmack, Michael Dell, Alexandr Wang, Palmer Luckey, Lisa Su, Jeff Dean, Fei-Fei Li, Paul Graham, Tom Brown, Kanjun Qiu, Joe Gebbia.

WATCH LIST

Added Aaron Levie (@levie), co-founder and CEO of Box. The list now stands at 45.

SILENT · 18

Marc Andreessen, Jensen Huang, Sam Altman, Demis Hassabis, Dario Amodei, Vitalik Buterin, Greg Brockman, Andrew Ng, Jim Fan, Mira Murati, Mark Zuckerberg, David Singleton, Ilya Sutskever, Sundar Pichai, Nat Friedman, Arthur Mensch, Guillaume Lample, Mark Chen.

Also on the wire

Alexandr Wang @alexandr_wang, chief AI officer at Meta, after an interview with David Senra about the Muse assistant: "kinda came up with this on the spot but might be my life motto now: give everyone the adulthood they dreamed of in childhood." Tobi Lütke: "Fantastic framing."

Tobi Lütke @tobi: "It's feeling really magical inside of Shopify right now. It's clearly the greatest time in the history of tech industry right now."

Guillermo Rauch @rauchg: "All plausible science fiction will be realized within our lifetime."

François Chollet @fchollet has been running large language model (LLM) post-training experiments with KerasHub, JAX and Google's tensor processing units (TPUs), and "Might open-source some of that." He also flagged Busabase, an open-source shared database for agents, because agent work today feels "very single-player."

Clément Delangue @ClementDelangue said Hugging Face Hub usage "is growing nicely" by daily active users, even as Nathan Lambert flagged falling downloads for top open models.

Uncle Bob Martin @unclebobmartin: "Agents are Aladdin's lamp, with a lot more than three wishes."

Elon Musk @elonmusk on an Andreessen Horowitz (a16z) chart showing call-center jobs shrinking 4 percent a year after 15 years of growth: "They will disappear fast due to SI."

Fei-Fei Li @drfeifei cheered a filmmaker who used World Labs' Atlas to make three new camera angles from a single image.

Kanjun Qiu @kanjun shared a founder's plan to run his business on Imbue Studio and compare it with Grok Bot.

Paul Graham @paulg on why AI is "kinda mid at writing," as Matthew Yglesias put it: "Apparently it's mostly just because they haven't made much of an effort to train it to write well yet."

The SI Sentinel is compiled from public posts on X and published reporting. Quotes are reproduced verbatim unless marked as translated or paraphrased.